

Troy (TJ) Collamore

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EDUCATION

Cornell University

Major: Computer Science, College of Engineering

Ithaca, NY

May 2030

Westford Academy

High Honors: 2022, 2023, 2024, 2025, 2026

Book Award: Clarkson University Leadership & Achievement Book Award 2025

Affiliations: German National Honor Society (2024 - 2026)

Majors of Interest: Computer Science, Engineering

Westford, MA

June 2026

Tufts University Pre-College Engineering with Artificial Intelligence Program

Summer 2025

- Built projects integrating mechanical systems, sensors, and machine learning algorithms, including:
 - Chronic Kidney Disease visualization using Nearest Neighbors algorithm
 - Race Car Simulation utilizing Reinforcement Learning & Artificial Neural Networking

Raytheon, Job Shadow

Spring 2025

- Selected to work with Chief Engineer Michael P. Clark

PROJECTS (featured at tjcollamore.com)

Blackjack Game

Summer 2025

- Designed and implemented in Java, focusing on independent mastery of object-oriented design patterns and recursion
- Purposefully limited AI assistance to UI elements only
- Game includes an 8-deck shoe, card shuffling, hit/stay/double/split/surrender logic, betting, dealer play, win detection, and bankroll
- **Skills:** Java, JavaFX, Replit, Game Logic, GUI Design, Object-Oriented Design, Recursion, Independent Research

Race Car Simulation, Tufts University Pre-College Program

Summer 2025

- Developed a race car simulation where a neural network learns optimal steering and acceleration using reinforcement learning
- Tuned reward functions and visualized model convergence throughout the training process
- **Skills:** Python, Claude Code, Reinforcement Learning, Neural Networks, Data Visualization

Pac-Man, MIT Scratch

Spring 2023

- Programmed a Pac-Man clone with multi-level progression, ghost behavior algorithms, and custom map design
- 3 unique levels with escalating speed; two-player mode where Player 2 overrides ghost behavior using keyboard controls
- **Skills:** Scratch, Game Logic, UI Design, Animations

Tar Spot Image Analysis, AP Biology Lab

Fall 2022

- Wrote a C# image-processing program to quantify how fungal tar spot affects chlorophyll concentration in leaves
- Combined biological analysis with computational modeling to enhance accuracy of chlorophyll measurement
- **Skills:** C#, Image Processing, Data Analysis, Computational Modeling, Biology

TRACK / CROSS COUNTRY

Westford Academy Spring and Winter Track & Field, Captain

2022 - 2026

- Main events: 2-mile, 1-mile, 800m
- **Personal Records: 9:51 2-mile, 4:44 1-mile, 2:03 800m**
- 8th-fastest 2-mile in school history
- Led a team of ~70 runners through training & warm-ups
- **Created & distributed training plans**, recommending training times, splits, and distances for each running group
 - Inspired by those of Stanford University & University of Colorado D1 teams via collaboration with WA alumni
- Commitment: 15 hours/week, 45 miles/week, 28 weeks/year

Westford Academy Cross Country (WA XC), Captain

2022 - 2026

- Competed in all DCL meets, invitationals, and championships
- **Personal Record: 16:47 5k**
- Selected to represent WA XC at the UMass Lowell Leadership Conference
 - Met with MA governors, professional runners, & NFL athletes
- Commitment: 15 hours/week, 45 miles/week, 14 weeks/year

COMMUNITY SERVICE

Nitehawks Track & Field Assistant Coach

Summer 2024

- Explained techniques and proper diet to elementary and middle school runners

Mass General Brigham Hospital Sports Volunteer

Summer 2024

- Modeled strengthening, stretching, and dynamic warm-up routines for youth athletes, to be used by doctors & physical therapists

Math Tutor

Summer 2024

- Prepared students for Westford Academy Algebra 2 Honors and Trig & Pre Calculus Honors material

Baystate Marathon Volunteer

2023 - 2024

- Prepared course, assigned bibs, and distributed drinks

Westford Flag Football Head Coach

Fall 2023

- Led a middle school flag football team to semi-finals
- Received the **League Sportsmanship Coaching Award**

EMPLOYMENT

Westford Flag Football Referee

2022 - 2023

- Refereed high school, middle school, and elementary school games for the Westford Flag Football League

Babysitter

2023 - 2025

- Cared for children, young teenagers, and pets